
Where the Admissible Radius Binds: a Correction and a Measured Boundary in Online Conformal Prediction

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Abstract

1 A printed claim that the scorecaster breaks online conformal coverage meets its
2 refutation by overlapping authors; this paper joins them and locates the edge
3 of the set that refutation relies on. Conformal PID states long-run coverage for
4 any bounded scorecaster; a corollary keeps it for predictable perturbations of the
5 deployed threshold inside an admissible radius. The claim was acted on; placement
6 and derivation are conceded by name. At the null scorecaster that condition is *tight*.
7 A legal L_1 dead band on the *completed* threshold leaves that set at its radius, and
8 past it miscoverage is 1.000000 against $\alpha = 0.1$. Coverage goes, not only the rate.
9 The edge belongs to the saturator–scorecaster pair, not the readout; elsewhere the
10 condition is sufficient only. Under the equally legal $\hat{q} \equiv +b/2$ the failing band
11 covers; at $\hat{q} \equiv -b/2$ both it and partial adjustment at $w = 0.999$ miscover.

1 Introduction

13 A refereed paper states that a scorecaster breaks Conformal PID’s coverage guarantee. A later one,
14 by the same first and last authors, proves in the same notation that a bounded perturbation of the
15 deployed threshold does not. The later cites the earlier, so this is ordinary self-correction, and the
16 joining is what this paper adds.

17 An online conformal method [Angelopoulos and Bates, 2021] emits a threshold before each obser-
18 vation and moves it. Forecast stability names, scores and prices that movement [Tunc et al., 2013,
19 Godahewa et al., 2025, Van Belle et al., 2023, Pritularga and Kourentzes, 2024, Genov et al., 2026,
20 Van Belle et al., 2026]. Varying temporal structure at fixed level came earlier, in forecast verification,
21 matching predictive marginals on real producers [Gneiting et al., 2007, Pinson and Girard, 2012,
22 Worsnop et al., 2018]. Coverage and mean length under-describe a deployed interval [Min et al.,
23 2026, Vaze, 2026]. None of it is claimed here.

24 **The record carries a claim and its refutation; this paper joins them.** Conformal PID [An-
25 gelopoulos et al., 2023] adds a *scorecaster* $\hat{q}_{t+1}$ to a saturating error integrator, asking only that it be
26 predictable and lie in $[-b/2, b/2]$: “any function of the past” (Theorem 1). Such a readout is already
27 quantified over. The claim reads “[s]ince using a scorecaster (D-part) in C-PID breaks the theoretical
28 coverage guarantee” [Li and Rodríguez, 2025], stated and acted on in its baselines paragraph, ver-
29 batim on refereed printed page 18443. It is refuted twice, with proof. Li et al. [2025, Corollary 2]
30 add any predictable d_t , $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ with $\mu_t \in [0, 1]$ predictable, to the deployed threshold;
31 since $\hat{q}_t + d_t$ still lies in $[-b/2, b/2]$ it is itself a legal scorecaster, so the result “has exactly the CPID
32 error-integration form” and $|E_T| = o(T)$. The same generic-slot argument carries AcMCP’s long-run
33 coverage [Wang and Hyndman, 2024]. Both identifiers and “scorecaster” were searched across arXiv
34 full text, arXiv abstracts, OpenReview, and Semantic Scholar’s six citers of Li and Rodríguez [2025]:
35 none returns both the claim and the corollary. Hu et al. [2026] still call scorecaster choice “arbitrary”
36 and lacking “principled guidance” while asserting valid coverage: model selection, not validity. **We**
37 **claim neither the placement nor the derivation.**

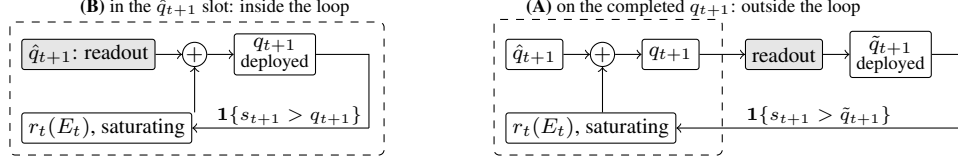


Figure 1: One scalar readout (grey) in its two places in (1); dashed is the loop r_t closes. In (B) the loop closes on the integrator’s own output, and both readouts keep $\{\hat{q}_t\}$ in the convex hull of $\hat{q}_1$ and the raw scorecaster’s range, inside Corollary 2’s admissible set. In (A) it closes on $\tilde{q}_{t+1}$, which it never computed, and the deployed sequence leaves that set.

38 **Where that refutation stops, the condition is tight.** Corollary 2 is a *retention* result, silent outside
 39 its radius, and the question it leaves is whether that radius is sharp. Section 3 exhibits a legal
 40 perturbation leaving it: the L_1 dead-band family characterised here on the *completed* threshold,
 41 whose edge is the corollary’s radius at the null scorecaster. Past it coverage goes, not only the rate: the
 42 published sufficient condition is necessary there. Section 4 sets both claims beside four vocabularies
 43 for the same movement.

44 2 Setup

45 **The producer.** A forecaster fit before t induces a score s_t , and a threshold q_t is emitted before
 46 y_t is revealed. The deployed set is $C_t = \{y : s_t(x_t, y) \leq q_t\}$ and $\text{err}_t = \mathbf{1}\{s_t(x_t, y_t) > q_t\}$. The
 47 target is long-run coverage, $T^{-1} \sum_{t \leq T} \text{err}_t = \alpha + o(1)$, with no model on the data. Conformal PID
 48 [Angelopoulos et al., 2023] manipulates the threshold on the score scale, where adaptive conformal
 49 inference [Gibbs and Candès, 2021] and DtACI [Gibbs and Candès, 2024] move the level,

$$q_{t+1} = \hat{q}_{t+1} + r_t \left(\sum_{i \leq t} (\text{err}_i - \alpha) \right), \quad (1)$$

50 their iteration (5). Its r_t saturates: condition (4) requires $|r_t(x)| \geq b$ with the sign of x once $|x| \geq$
 51 $c h(t)$, for h nonnegative, nondecreasing and sublinear. With $E_t = \sum_{i \leq t} (\text{err}_i - \alpha)$, Proposition 2
 52 gives $|E_T| \leq c h(T) + 1$, read on realised scores.

53 **One scalar, two readouts, two places.** Damping a sequentially updated scalar has two standard
 54 readouts, both prior work. A quadratic cost gives linear partial adjustment $\tilde{q}_{t+1} = (1 - \lambda)\hat{q}_{t+1}^{\text{raw}} + \lambda\tilde{q}_t$
 55 [Gârleanu and Pedersen, 2013], published as post-processing [Godahewa et al., 2025] and applied
 56 to a deployed threshold [Binny and Dixit, 2025, Eq. 13]. A proportional cost gives a no-trade band,
 57 moving only when the target is more than τ away, then pulling back exactly τ [Constantinides,
 58 1986, Davis and Norman, 1990]. Either readout has two positions in (1), drawn in Figure 1; (A) is
 59 measured here. *The dead band (L_1) is not a stylistic alternative to the partial-adjustment (L_2) form;*
 60 *it is the family whose failure Section 3 characterises. Neither is safe by construction: L_2 forfeits*
 61 *Proposition 2’s rate while covering, L_1 past the radius loses coverage, and at the legal $\hat{q} \equiv -b/2$ both*
 62 *lose it. What is tight is the radius, not a form.*

63 **The quantity, and the names it already has.** We report the deployed threshold’s *travel*, $\mathcal{T}_H =$
 64 $\sum_{t=2}^H |q_t - q_{t-1}|$, after actuator travel in process control. The arithmetic is not new. Over a predictive
 65 distribution $\sum |\Delta q|$ is the Multi-Quantile Change [Zanotti, 2026, v3, §3.4.1]; verification calls it
 66 *jumpiness*, a *convergence index*, (in)consistency [Zsótér et al., 2009, Ehret, 2010, Pappenberger
 67 et al., 2011]; applied control, IACER [Sarhadi, 2025]; dynamic-regret online learning, a path length.
 68 Table 2’s caption names ten, so a fresh one is used here.

69 **Harness.** The harness fixes $\alpha = 0.1$ and $b = 2$. Its saturator $r_t(x) = b \text{clip}(x/(c h(t)), -1, 1)$,
 70 with $c = 1$ and $h(t) = \log(t + 2)$, meets (4) with equality. The scorecaster is null and legal, reducing
 71 (1) to the pure error integration Proposition 2 addresses. Neither choice is neutral: (4) is a lower
 72 bound met exactly here, and Theorem 1 permits scorecasters $b/2$ either side of null. The adversary is
 73 specified, not tie-broken. It plays $s_t = \text{clip}(\tilde{q}_t + \varepsilon, \pm b/2)$, $\varepsilon = 10^{-9}$, against the *deployed* $\tilde{q}_t$: the
 74 $\varepsilon \downarrow 0$ reading of “sets the score at the threshold”, firing the strict indicator. Legality is Theorem 1’s
 75 $[-b/2, b/2]$ on both sides, so $b/2 + b/2 = b$ is exactly tight.

Table 1: Placement A: a one-scalar readout on the *completed* threshold, under one deterministic adversary, on one pass to $T = 10^6$; all eleven arms run are listed. Proposition 2’s bound is 13.2061 at $T = 2 \times 10^5$ and 14.8155 at $T = 10^6$; “ratio” is $\max_t |E_t|$ over it, and “Sat.” the fraction of rounds on which (4) delivers its full $\pm b$. Every number is the *primary* regime’s, defined in Appendix A, which also gives the $T = 10^4$ excursions and the fitted excursion law.

Readout on the completed threshold	miscov. 2×10^5	miscov. 10^6	$\max_t E_t 10^6$	ratio	Sat. 10^6	travel 10^6
none (control)	0.100035	0.100007	7.80	0.53	0.0000	28,311
partial adj. $w = 0.5$	0.100030	0.100007	7.70	0.52	0.0000	18,113
partial adj. $w = 0.9$	0.100030	0.100007	7.50	0.51	0.0000	4,081
partial adj. $w = 0.99$	0.100040	0.100006	63.00	4.25	0.0007	1,023
partial adj. $w = 0.999$	0.100035	0.100004	623.70	42.10	0.0097	202
growing time constant $p = 0.5$	0.100030	0.100006	12.60	0.85	0.0000	330
growing time constant $p = 0.9$	0.100030	0.100024	53.00	3.58	0.3181	16
running mean	0.099940	0.100010	49.50	3.34	0.3647	9
dead band $\tau = 0.5$	0.100035	0.100005	10.60	0.72	0.0000	2,045
dead band $\tau = 0.9$	0.100060	0.100012	13.20	0.89	0.0038	1,051
dead band $\tau = 1.5$	1.000000	1.000000	900,000	60,747	1.0000	0.5

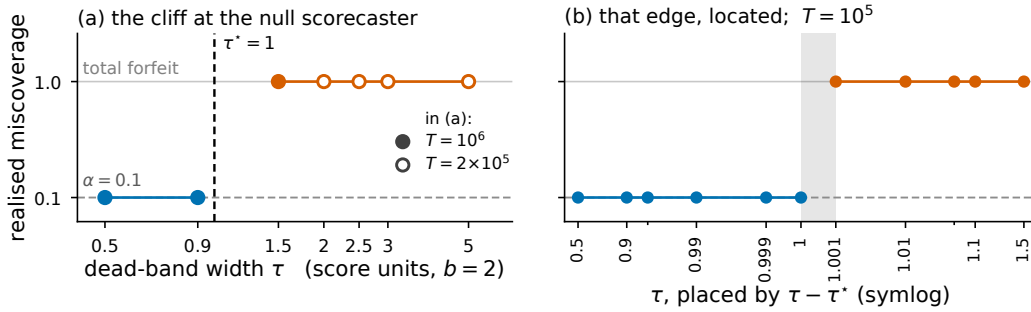


Figure 2: Realised miscoverage against dead-band width τ at the null scorecaster, $\hat{q} \equiv 0$ with a saturator of level b , where $\tau^* = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$ is 1. (a) The full sweep: filled markers run to $T = 10^6$ and the four widest bands, drawn open, to $T = 2 \times 10^5$; blue covers, orange forfeits, and the dashed rule is τ^* . (b) The same setting at $T = 10^5$ over eleven measured widths, placed by $\tau - \tau^*$ on a symlog axis whose linear zone is $|\tau - \tau^*| \leq 10^{-3}$, which magnifies the neighbourhood of τ^* : visual spacing here is not proportional to true distance in τ . Unlabelled minor ticks mark $\tau = 0.95$ and 1.05 . The grey band spans the widest covering width to the narrowest failing one; Figure 3 gives the other four $(r_t, \hat{q})$ settings.

3 The edge of the admissible set, measured

The retention result is published, and tested from outside. Li et al. [2025, Corollary 2] prove that a predictable d_t with $|d_t| \leq \mu_t(b/2 - |\hat{q}_t|)$ on the deployed threshold keeps (1) in error-integration form and retains $|E_T| \leq c h(T) + 1$: the radius is theirs. A downstream partial adjustment of weight w leaves it; Table 1 prices that departure, and five of its eleven arms exceed the bound at $T = 10^6$. For $w = 0.99$ and 0.999 the excursion is *flat* in T while the bound grows like $\log T$. At $w = 0.999$ the ratio is 61.08 at $T = 10^4$ and 42.10 at $T = 10^6$. The forfeit is a property of the smoothing *strength*, not Placement A, and it buys something: travel falls from 28,311 to 202. The forfeit’s sign is counter-intuitive. A readout on the completed q_{t+1} touches neither r_t nor the integrator’s argument, so it cannot put (4) at risk. It delays a correction already computed, and the accumulator excurses *further*. Smoothing the output does not damp E_t ; it feeds it. That 623.70 turns on the adversary’s ε , not the counting convention.

The L_1 dead band leaves the radius, and the edge is measurable. A dead band pulls the deployed threshold back by exactly τ when it releases, so “inside Corollary 2” reads $\tau \leq \mu_t(b/2 - |\hat{q}_t|)$. Under saturation the deployed value is pinned τ below the integrator’s ceiling, and its realised supremum is $b - \tau$ to the printed digit at every $\tau \geq 0.9$. The adversary’s best legal play is $b/2$, so coverage is lost when that supremum drops below it, at $\tau^* = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$. All 19 dead-band runs agree with that law (Figure 3); only the null scorecaster’s edge is located (Figure 2), the others one-sided. **Here** $\sup_x r_t(x) = b$ and $\hat{q} \equiv 0$, so $\tau^* = b/2$, one point of the admissible class, and both terms move it. The legal $\hat{q} \equiv +b/2$ carries the $\tau = 1.5$ band inside the bound. The equally

96 legal $\hat{q} \equiv -b/2$ drops the edge to 0, and there $\tau = 0.5$ returns 1.000000, as does partial adjustment
 97 at $w = 0.999$. Condition (4) bounds $\sup_x r_t(x)$ only from below, so raising it costs the theorem
 98 nothing; it moves the edge too (Figure 3) and shrinks the excursion. Under the unbounded tangent
 99 integrator Angelopoulos et al. [2023] report using in all their experiments, τ^* is unbounded and all
 100 three widths there cover. The excursion is 20.10 rather than 623.70, both at $T = 10^6$. **The method's**
 101 **own integrator therefore sits inside the set at every width run here.** Outside the set the transition
 102 is a step. Past τ^* the threshold sits permanently below the reachable score range and $|E_t| = (1 - \alpha)t$
 103 grows linearly, 60,747 times the bound at $T = 10^6$. The wider bands $\tau = 2.0, 2.5, 3.0$ and 5.0
 104 fail too at $T = 2 \times 10^5$ (Figure 2), so the failing set is (τ^*, ∞) , open at the left. **The endpoint is**
 105 **where the adversary's specification decides the answer.** At $\tau = \tau^*$ the deployed supremum equals
 106 the adversary's own cap $b/2$, so the clip forces an exact play and the reading settles it: the primary
 107 regime covers. Appendix A separates the two readings and prints both. **The failing arms have**
 108 **Sat. = 1.0000.** Condition (4) delivers its full $\pm b$ on every round, and what fails is Proposition 2's
 109 other hypothesis: the threshold closing the loop *is* the integrator's output.

110 **What that measures is that the published condition is tight.** At $\mu_t = 1$, $\hat{q} \equiv 0$ the measured
 111 τ^* is exactly Corollary 2's radius, and crossing it costs coverage rather than the rate. **The claim is**
 112 **that necessity: a tightness result about someone else's sufficient condition,** smaller and more
 113 precise than a forfeit of the rate by post-processing. Elsewhere it stays sufficient only: at $\hat{q} \equiv +b/2$
 114 Corollary 2 permits radius 0 while every width run there covers. That asymmetry makes "tight" a
 115 located claim.

116 **The constructive reading, and it is the corollary's rather than ours.** Placement B keeps $\{\hat{q}_t\}$
 117 inside the admissible radius, so Theorem 1 carries it already. What the measurement adds is that the
 118 band has an edge, computable before deployment from $\sup_x r_t(x)$ and $\sup_t \hat{q}_t$, and that crossing it is
 119 not a graceful degradation.

120 4 Where this sits

121 Four vocabularies name one quantity; Table 2 (App. B) names all four. **What is conceded, by**
 122 **name.** The placement is prior work: Angelopoulos et al. [2023] state coverage for any admissible
 123 integrator and any scorecaster, and run a smoothing-family model in that slot themselves. Dupuy
 124 et al. [2026, Appendix A] give the generic argument, and Duerst et al. [2024] adopt that scorecaster.
 125 The derivation is prior work too, and this concession matters most. Li et al. [2025, Corollary 2]
 126 prove the retention statement in Conformal PID's own notation, with Wang and Hyndman [2024] an
 127 independent second instance. Section 2 concedes the smoother object, the metric arithmetic and both
 128 readout maps. Zheng et al. [2025], Wu et al. [2025], Chen et al. [2023], Lade et al. [2026] each place a
 129 smoother inside their own validity argument. One vocabulary out, this is integrator anti-windup with a
 130 feedforward path: free-until-the-bound-binds is its defining specification, not a finding [Galeani et al.,
 131 2009]. Nearest in the other direction, Gao et al. [2025] run the same recursion but gate the *model*
 132 update on the conformal score. Their one non-standard assumption is that $\sum_{t \leq T} q_t \leq O(\alpha h(T))$
 133 for a sublinear h , supported with a plot rather than derived.

134 5 Limitations

135 **The evidence is one construction, one adversary, one saturator;** and $0.63/(1 - w)$ is specific to
 136 the saturator's *level*, which condition (4) bounds only from below. **The inherited guarantee is weak**
 137 **where it applies.** Under the tangent integrator $h(t) = t/\log t$, the certified gap $(ch(T) + 1)/T$
 138 decays as $O(1/\log T)$ rather than $O(1/T)$. What is forfeited was not a tight certificate. **The dead-**
 139 **band widths are absolute, and the failure is total only against the adversary.** τ is in score units
 140 with $b = 2$, so the failing $\tau = 1.5$ is $0.75b$. Under i.i.d. scores it returns 0.249376 at $T = 10^6$, an
 141 adversary-free confirmation of the mechanism rather than a retreat. The threshold pins at $b - \tau = 0.5$,
 142 and uniform scores on $[-b/2, b/2]$ put $P(s > 0.5) = 0.25$ exactly. **The harness was rebuilt**
 143 **against numbers already read,** which carries less weight than a rerun done without sight of them.
 144 **Placement B is conceded, not tested. The i.i.d. check above used one seed** (20260820, $T = 10^6$);
 145 **the primary, adversarial harness uses none, and only its null-scorecaster edge is bracketed;** the
 146 others are one-sided. **What would overturn the claim** is a legal $(r_t, \hat{q})$ pair with $\tau \leq \tau^*$ that loses
 147 coverage, or $\tau > \tau^*$ that keeps it; the sweep has neither.

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241 Code and data

242 The simulator, its frozen configuration, the results/ files behind every figure and table, the gen-
 243 erators that read them, and this project’s audit trail across sessions are at [https://\[REPOSITORY](https://[REPOSITORY LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION])
 244 LINK - INSERTED BY AUTHOR PRIOR TO SUBMISSION].

245 A Supporting measurements

246 **The primary regime.** Every number in Table 1 is measured under one deterministic adversary,
 247 which consumes no randomness and needs no seed. It plays $s_t = \text{clip}(\hat{q}_t + \varepsilon, \pm b/2)$ with $\varepsilon = 10^{-9}$
 248 against the *deployed* threshold, and the indicator is strict. The specification is not a tie rule, and
 249 the distinction is measured rather than argued. Scoring an exact-threshold play as a *miscoverage*
 250 reproduces every quantity Table 1 reports: 44 of 44 arm-horizon cells identical, all eleven arms at
 251 all four horizons. Scoring the same play under the *strict* indicator does not, and it moves every
 252 row, because the adversary’s signature move then covers rather than misses. Partial adjustment at
 253 $w = 0.999$ excurses 70.80 in place of 623.70, and the failing dead band returns 0.000000 in place of
 254 1.000000. The $\varepsilon \downarrow 0$ limit the harness specifies is the first reading. **No row is convention-free with**
 255 **respect to the second**, which is why the adversary is specified rather than tie-broken. At $\tau = \tau^*$
 256 itself the two readings part: the strict indicator returns 0.000000 where scoring the exact play as a
 257 miscoverage returns 1.000000.

258 **Excursions at $T = 10^4$, and where the excursion law holds.** Partial adjustment excurses 6.30,
 259 63.00 and 623.70 at $w = 0.9, 0.99$ and 0.999 . The last two are flat in T while Proposition 2’s
 260 bound grows like $\log T$, which is why the ratio falls with the horizon. On the control and the four
 261 partial-adjustment arms the excursion fits $\max_t |E_t| \approx \max\{0.5 h(T) + 0.9, 0.63/(1 - w)\}$ at every
 262 horizon, most loosely at $w = 0.9$, the one arm whose two branches cross inside the horizon range.
 263 This is a fit to those five arms, not a derived bound. It does not extend to the other six: the growing
 264 time constants, the running mean and the dead bands all excurse further than it predicts.

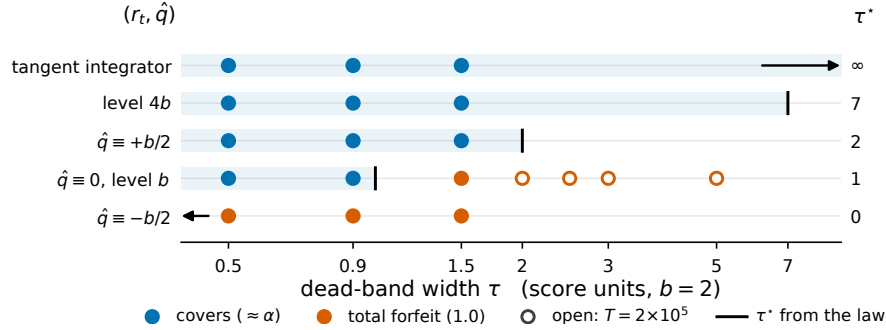


Figure 3: All 19 dead-band runs, at the five admissible $(r_t, \hat{q})$ settings: rows are settings ordered by τ^* , and the shared horizontal axis is τ . Blue markers cover at $\approx \alpha$ and orange markers return 1.000000; filled markers run to $T = 10^6$ and open markers to $T = 2 \times 10^5$. The rule on each row is that setting’s $\tau^* = \sup_x r_t(x) + \sup_t \hat{q}_t - b/2$, printed at the right, and the tinted band is $\tau \leq \tau^*$. Two rules fall off the axis and are drawn as arrows: $\tau^* = 0$ at $\hat{q} \equiv -b/2$, and τ^* unbounded under Angelopoulos et al.’s tangent integrator.

265 **Realised miscoverage at the covering settings of Figure 3.** At $\hat{q} \equiv +b/2$ the $\tau = 1.5$ band returns
 266 0.100010 with $\max_t |E_t| = 11.20$, inside the bound. Under Angelopoulos et al.’s tangent integrator
 267 the same width returns 0.100001. Raising the saturator’s level to $4b$ leaves coverage intact and cuts
 268 the $w = 0.999$ excursion from 623.70 to 120.60. That is the first term of τ^* moving the edge, not the
 269 readout changing behaviour.

B The four vocabularies

Table 2: Four literatures name the movement of a sequentially updated forecast in four vocabularies: *abrupt threshold changes*; *forecast (in)stability* (MASC, MAC/SDC, MQC/SMQC, congruence, nervousness); *jumpiness*, *convergence index*, *(in)consistency*; and *switching cost*. The row gives what each contributes.

Online conformal	Forecast stability	Verification	Switching-cost online learning
a distribution-free coverage guarantee for any bounded scorecaster, with a rate [Angelopoulos et al., 2023, Thm 1, Prop. 2], and the admissible set measured here to be tight	long-run- the scored movement functional, the readout map as post-processing [Godaheva et al., 2025, Eq. 3], and a priced decision [Genov et al., 2026, §4.2] and [Van Belle et al., 2026]	the matched-level design on real producers from 2012, with a control <i>stronger</i> than the pair: the whole marginal [Pinson and Girard, 2012, Worsnop et al., 2018]	worst-case theory for the penalised problem: Thm 2, and a Thm 7 trade-off in which sublinear regret is bought by growing θ with T , which grows the ratio [Andrew et al., 2013]